

Newman-Penrose Formalism Calculations

Stephani et al., Equation (24.22)

Metric

$$\begin{aligned}
 ds^2 = & \frac{4(3\Lambda l^6 - 6\Lambda l^4 x^2 - \Lambda l^2 x^4 + 6l^2 nx - 3(e^2 + g^2)l^2)y^2}{3(l^2 + x^2)} du \otimes du + \frac{2(3\Lambda l^5 - 6\Lambda l^3 x^2 - \Lambda l x^4 + 6lnx - 3(e^2 + g^2)l)y}{3(l^2 + x^2)} du \otimes dv + (l^2 + x^2) du \\
 & \otimes dy + \frac{2(3\Lambda l^5 - 6\Lambda l^3 x^2 - \Lambda l x^4 + 6lnx - 3(e^2 + g^2)l)y}{3(l^2 + x^2)} dv \otimes du + \left(\frac{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}{3(l^2 + x^2)} \right) dv \\
 & \otimes dv + \left(\frac{3(l^2 + x^2)}{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx} \right) dx \otimes dx + (l^2 + x^2) dy \otimes du
 \end{aligned}$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (-1) du$$

$$n = \left(\frac{2\Lambda l^6 y^2}{l^2 + x^2} - \frac{4\Lambda l^4 x^2 y^2}{l^2 + x^2} - \frac{2\Lambda l^2 x^4 y^2}{3(l^2 + x^2)} - \frac{2e^2 l^2 y^2}{l^2 + x^2} - \frac{2g^2 l^2 y^2}{l^2 + x^2} + \frac{4l^2 nxy^2}{l^2 + x^2} \right) du + \left(\frac{2\Lambda l^5 y}{l^2 + x^2} - \frac{4\Lambda l^3 x^2 y}{l^2 + x^2} - \frac{2\Lambda l x^4 y}{3(l^2 + x^2)} - \frac{2e^2 l y}{l^2 + x^2} - \frac{2g^2 l y}{l^2 + x^2} + \frac{4lnxy}{l^2 + x^2} \right) dv + (l^2 + x^2) dy$$

$$m = \left(\sqrt{\frac{1}{6}} \sqrt{\frac{3\Lambda l^4}{l^2 + x^2} - \frac{6\Lambda l^2 x^2}{l^2 + x^2} - \frac{\Lambda x^4}{l^2 + x^2} - \frac{3e^2}{l^2 + x^2} - \frac{3g^2}{l^2 + x^2} + \frac{6nx}{l^2 + x^2}} \right) dv + \left(\frac{3i\sqrt{\frac{2}{3}}}{2\sqrt{\frac{3\Lambda l^4}{l^2 + x^2} - \frac{6\Lambda l^2 x^2}{l^2 + x^2} - \frac{\Lambda x^4}{l^2 + x^2} - \frac{3e^2}{l^2 + x^2} - \frac{3g^2}{l^2 + x^2} + \frac{6nx}{l^2 + x^2}}} \right) dx$$

$$\bar{m} = \left(\sqrt{\frac{1}{6}} \sqrt{\frac{3\Lambda l^4}{l^2 + x^2} - \frac{6\Lambda l^2 x^2}{l^2 + x^2} - \frac{\Lambda x^4}{l^2 + x^2} - \frac{3e^2}{l^2 + x^2} - \frac{3g^2}{l^2 + x^2} + \frac{6nx}{l^2 + x^2}} \right) dv + \left(-\frac{3i\sqrt{\frac{2}{3}}}{2\sqrt{\frac{3\Lambda l^4}{l^2 + x^2} - \frac{6\Lambda l^2 x^2}{l^2 + x^2} - \frac{\Lambda x^4}{l^2 + x^2} - \frac{3e^2}{l^2 + x^2} - \frac{3g^2}{l^2 + x^2} + \frac{6nx}{l^2 + x^2}}} \right) dx$$

Contravariant components

$$l = \left(-\frac{1}{l^2 + x^2} \right) dy$$

$$n = (1) du + \left(-\frac{2\Lambda l^6 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{4\Lambda l^4 x^2 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{2\Lambda l^2 x^4 y^2}{3(l^4 + 2l^2 x^2 + x^4)} + \frac{2e^2 l^2 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{2g^2 l^2 y^2}{l^4 + 2l^2 x^2 + x^4} - \frac{4l^2 n x y^2}{l^4 + 2l^2 x^2 + x^4} \right) dy$$

$$\begin{aligned} m = & \left(\frac{\sqrt{3}\sqrt{2}l^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} + \frac{\sqrt{3}\sqrt{2}x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} \right) dv \\ & + \left(\frac{3i\sqrt{l^2 + x^2}\Lambda l^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} - \frac{6i\sqrt{l^2 + x^2}\Lambda l^2 x^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \right. \\ & \quad - \frac{i\sqrt{l^2 + x^2}\Lambda x^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} - \frac{3i\sqrt{l^2 + x^2}e^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & \quad \left. - \frac{3i\sqrt{l^2 + x^2}g^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} + \frac{6i\sqrt{l^2 + x^2}nx}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \right) dx \\ & + \left(-\frac{\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}ly}{(\sqrt{3}l^2 + \sqrt{3}x^2)\sqrt{l^2 + x^2}} \right) dy \end{aligned}$$

$$\begin{aligned} \bar{m} = & \left(\frac{\sqrt{3}\sqrt{2}l^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} + \frac{\sqrt{3}\sqrt{2}x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} \right) dv \\ & + \left(-\frac{3i\sqrt{l^2 + x^2}\Lambda l^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} + \frac{6i\sqrt{l^2 + x^2}\Lambda l^2 x^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \right. \\ & \quad + \frac{i\sqrt{l^2 + x^2}\Lambda x^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} + \frac{3i\sqrt{l^2 + x^2}e^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & \quad \left. + \frac{3i\sqrt{l^2 + x^2}g^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} - \frac{6i\sqrt{l^2 + x^2}nx}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \right) dx \\ & + \left(-\frac{\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}ly}{(\sqrt{3}l^2 + \sqrt{3}x^2)\sqrt{l^2 + x^2}} \right) dy \end{aligned}$$

Directional Derivatives

$$D = -\frac{1}{l^2 + x^2} \partial_y$$

$$\Delta = \partial_u - \frac{2\Lambda l^6 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{4\Lambda l^4 x^2 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{2\Lambda l^2 x^4 y^2}{3(l^4 + 2l^2 x^2 + x^4)} + \frac{2e^2 l^2 y^2}{l^4 + 2l^2 x^2 + x^4} + \frac{2g^2 l^2 y^2}{l^4 + 2l^2 x^2 + x^4} - \frac{4l^2 n x y^2}{l^4 + 2l^2 x^2 + x^4} \partial_y$$

$$\begin{aligned} \delta = & \frac{\sqrt{3}\sqrt{2}l^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} + \frac{\sqrt{3}\sqrt{2}x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} \partial_v \\ & + \frac{3i\sqrt{l^2 + x^2}\Lambda l^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & - \frac{6i\sqrt{l^2 + x^2}\Lambda l^2 x^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} - \frac{i\sqrt{l^2 + x^2}\Lambda x^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & - \frac{3i\sqrt{l^2 + x^2}e^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} - \frac{3i\sqrt{l^2 + x^2}g^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & + \frac{6i\sqrt{l^2 + x^2}nx}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \partial_x - \frac{\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}ly}{(\sqrt{3}l^2 + \sqrt{3}x^2)\sqrt{l^2 + x^2}} \partial_y \end{aligned}$$

$$\begin{aligned} \bar{\delta} = & \frac{\sqrt{3}\sqrt{2}l^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} + \frac{\sqrt{3}\sqrt{2}x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}} \partial_v \\ & - \frac{3i\sqrt{l^2 + x^2}\Lambda l^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} + \frac{6i\sqrt{l^2 + x^2}\Lambda l^2 x^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & + \frac{i\sqrt{l^2 + x^2}\Lambda x^4}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & + \frac{3i\sqrt{l^2 + x^2}e^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} + \frac{3i\sqrt{l^2 + x^2}g^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \\ & - \frac{6i\sqrt{l^2 + x^2}nx}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)} \partial_x - \frac{\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}ly}{(\sqrt{3}l^2 + \sqrt{3}x^2)\sqrt{l^2 + x^2}} \partial_y \end{aligned}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{8i\,\Lambda l^5xy}{l^4+2\,l^2x^2+x^4} + \frac{8i\,\Lambda l^3x^3y}{3\,(l^4+2\,l^2x^2+x^4)} + \frac{2i\,l^3ny}{l^4+2\,l^2x^2+x^4} + \frac{4i\,e^2lxy}{l^4+2\,l^2x^2+x^4} + \frac{4i\,g^2lxy}{l^4+2\,l^2x^2+x^4} - \frac{6i\,lnx^2y}{l^4+2\,l^2x^2+x^4}$$

$$\lambda = 0$$

$$\begin{aligned}
\nu = & \frac{6\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^{11}y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} - \frac{24i\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^{10}xy^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& - \frac{24\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^9x^2y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{56i\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^8x^3y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{20\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^7x^4y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& - \frac{8i\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^6x^5y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{8\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^5x^6y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& - \frac{8i\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^4x^7y^2}{3(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{2\sqrt{2}\sqrt{l^2+x^2}\Lambda^2l^3x^8y^2}{3(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& - \frac{12\sqrt{2}\sqrt{l^2+x^2}\Lambda e^2l^7y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& - \frac{12\sqrt{2}\sqrt{l^2+x^2}\Lambda g^2l^7y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{6i\sqrt{2}\sqrt{l^2+x^2}\Lambda l^8ny^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{36i\sqrt{2}\sqrt{l^2+x^2}\Lambda e^2l^6xy^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{36i\sqrt{2}\sqrt{l^2+x^2}\Lambda g^2l^6xy^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{24\sqrt{2}\sqrt{l^2+x^2}\Lambda l^7nxy^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{24\sqrt{2}\sqrt{l^2+x^2}\Lambda e^2l^5x^2y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} \\
& + \frac{24\sqrt{2}\sqrt{l^2+x^2}\Lambda g^2l^5x^2y^2}{(\sqrt{3}l^6+3\sqrt{3}l^4x^2+3\sqrt{3}l^2x^4+\sqrt{3}x^6)\sqrt{3\Lambda l^4-6\Lambda l^2x^2-\Lambda x^4-3e^2-3g^2+6nx}} - \dots
\end{aligned}$$

$$\begin{aligned}
\alpha = & \frac{\sqrt{3}\sqrt{2}\Lambda l^5}{4\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} - \frac{i\sqrt{3}\sqrt{2}\Lambda l^4 x}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} \\
& - \frac{\sqrt{3}\sqrt{2}\Lambda l^3 x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} - \frac{2i\sqrt{3}\sqrt{2}\Lambda l^2 x^3}{3\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} \\
& - \frac{\sqrt{3}\sqrt{2}\Lambda l x^4}{12\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} \\
& - \frac{i\sqrt{3}\sqrt{2}\Lambda x^5}{6\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} - \frac{\sqrt{3}\sqrt{2}e^2 l}{4\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} \\
& - \frac{\sqrt{3}\sqrt{2}g^2 l}{4\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} + \frac{i\sqrt{3}\sqrt{2}l^2 n}{4\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} \\
& + \frac{\sqrt{3}\sqrt{2}l n x}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}} + \frac{i\sqrt{3}\sqrt{2}n x^2}{4\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(l^2 + x^2)^{\frac{3}{2}}}
\end{aligned}$$

$$\begin{aligned}
\beta = & \frac{3\Lambda l^5}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} - \frac{6i\Lambda l^4 x}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} \\
& - \frac{3\Lambda l^3 x^2}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} + \frac{2i\Lambda l^2 x^3}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} \\
& - \frac{\Lambda l x^4}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} - \frac{3e^2 l}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} \\
& - \frac{3g^2 l}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} + \frac{3i l^2 n}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} \\
& + \frac{3i e^2 x}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} + \frac{3i g^2 x}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} \\
& + \frac{3l n x}{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}} - \frac{9i n x^2}{2\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}(\sqrt{3}\sqrt{2}l^2 + \sqrt{3}\sqrt{2}x^2)\sqrt{l^2 + x^2}}
\end{aligned}$$

$$\tau = \frac{\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}l}{\sqrt{3}\sqrt{2}l^4 + 2\sqrt{3}\sqrt{2}l^2 x^2 + \sqrt{3}\sqrt{2}x^4} - \frac{i\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}x}{\sqrt{3}\sqrt{2}l^4 + 2\sqrt{3}\sqrt{2}l^2 x^2 + \sqrt{3}\sqrt{2}x^4}$$

$$\pi = -\frac{\sqrt{3}\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}l}{6(l^4 + 2l^2 x^2 + x^4)} - \frac{i\sqrt{3}\sqrt{2}\sqrt{3\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}\sqrt{l^2 + x^2}x}{6(l^4 + 2l^2 x^2 + x^4)}$$

$$\epsilon = 0$$

$$\begin{aligned} \gamma = & \frac{2\Lambda l^6 y}{l^4 + 2l^2 x^2 + x^4} - \frac{4i\Lambda l^5 xy}{l^4 + 2l^2 x^2 + x^4} - \frac{4\Lambda l^4 x^2 y}{l^4 + 2l^2 x^2 + x^4} + \frac{4i\Lambda l^3 x^3 y}{3(l^4 + 2l^2 x^2 + x^4)} - \frac{2\Lambda l^2 x^4 y}{3(l^4 + 2l^2 x^2 + x^4)} - \frac{2e^2 l^2 y}{l^4 + 2l^2 x^2 + x^4} \\ & - \frac{2g^2 l^2 y}{l^4 + 2l^2 x^2 + x^4} + \frac{i l^3 n y}{l^4 + 2l^2 x^2 + x^4} + \frac{2i e^2 l x y}{l^4 + 2l^2 x^2 + x^4} + \frac{2i g^2 l x y}{l^4 + 2l^2 x^2 + x^4} + \frac{4l^2 n x y}{l^4 + 2l^2 x^2 + x^4} - \frac{3i l n x^2 y}{l^4 + 2l^2 x^2 + x^4} \end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{e^2}{2(l^4 + 2l^2 x^2 + x^4)} + \frac{g^2}{2(l^4 + 2l^2 x^2 + x^4)}$$

$$\begin{aligned}
\Phi_{12} = & \frac{6\sqrt{l^2+x^2}\Lambda e^2 l^5 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{6\sqrt{l^2+x^2}\Lambda g^2 l^5 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}\Lambda e^2 l^3 x^2 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}\Lambda g^2 l^3 x^2 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{2\sqrt{l^2+x^2}\Lambda e^2 l x^4 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{2\sqrt{l^2+x^2}\Lambda g^2 l x^4 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{6\sqrt{l^2+x^2}e^4 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}e^2 g^2 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{6\sqrt{l^2+x^2}g^4 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{12\sqrt{l^2+x^2}e^2 l n x y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{12\sqrt{l^2+x^2}g^2 l n x y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}}
\end{aligned}$$

$$\Phi_{20} = 0$$

$$\begin{aligned}
\Phi_{21} = & \frac{6\sqrt{l^2+x^2}\Lambda e^2 l^5 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{6\sqrt{l^2+x^2}\Lambda g^2 l^5 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}\Lambda e^2 l^3 x^2 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}\Lambda g^2 l^3 x^2 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{2\sqrt{l^2+x^2}\Lambda e^2 l x^4 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{2\sqrt{l^2+x^2}\Lambda g^2 l x^4 y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{6\sqrt{l^2+x^2}e^4 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{12\sqrt{l^2+x^2}e^2 g^2 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& - \frac{6\sqrt{l^2+x^2}g^4 l y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{12\sqrt{l^2+x^2}e^2 l n x y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}} \\
& + \frac{12\sqrt{l^2+x^2}g^2 l n x y}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4 x^2 + 3\sqrt{3}\sqrt{2}l^2 x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{3}\Lambda l^4 - 6\Lambda l^2 x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx}}
\end{aligned}$$

$$\begin{aligned}
\Phi_{22} = & \frac{4\Lambda e^2 l^6 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} + \frac{4\Lambda g^2 l^6 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} - \frac{8\Lambda e^2 l^4 x^2 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} - \frac{8\Lambda g^2 l^4 x^2 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} - \frac{4\Lambda e^2 l^2 x^4 y^2}{3(l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6)} \\
& - \frac{4\Lambda g^2 l^2 x^4 y^2}{3(l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6)} - \frac{4e^4 l^2 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} - \frac{8e^2 g^2 l^2 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} - \frac{4g^4 l^2 y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} + \frac{8e^2 l^2 n x y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6} + \frac{8g^2 l^2 n x y^2}{l^6 + 3l^4 x^2 + 3l^2 x^4 + x^6}
\end{aligned}$$

$$\Lambda = \frac{1}{6} \Lambda$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & -\frac{4\Lambda l^6}{3(l^6 + 3l^4x^2 + 3l^2x^4 + x^6)} + \frac{4i\Lambda l^5x}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} + \frac{4\Lambda l^4x^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{4i\Lambda l^3x^3}{3(l^6 + 3l^4x^2 + 3l^2x^4 + x^6)} \\ & + \frac{e^2l^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} + \frac{g^2l^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{il^3n}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{2ie^2lx}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{2ig^2lx}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} \\ & - \frac{3l^2nx}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{e^2x^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} - \frac{g^2x^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} + \frac{3ilnx^2}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6} + \frac{nx^3}{l^6 + 3l^4x^2 + 3l^2x^4 + x^6}\end{aligned}$$

$$\begin{aligned}\Psi_3 = & -\frac{8\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx\Lambda l^7y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} + \frac{24i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx\Lambda l^6xy}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & + \frac{24\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx\Lambda l^5x^2y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} - \frac{8i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nx\Lambda l^4x^3y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & + \frac{6\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxe^2l^3y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} + \frac{6\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxg^2l^3y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & - \frac{6i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxl^4ny}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} - \frac{12i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxe^2l^2xy}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & - \frac{12i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxg^2l^2xy}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} - \frac{18\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxl^3nxy}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & - \frac{6\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxe^2lx^2y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} - \frac{6\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxg^2lx^2y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} \\ & + \frac{18i\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxl^2nx^2y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}} + \frac{6\sqrt{3\Lambda l^4 - 6\Lambda l^2x^2 - \Lambda x^4 - 3e^2 - 3g^2 + 6nxl^2nx^3y}}{(\sqrt{3}\sqrt{2}l^6 + 3\sqrt{3}\sqrt{2}l^4x^2 + 3\sqrt{3}\sqrt{2}l^2x^4 + \sqrt{3}\sqrt{2}x^6)\sqrt{l^2 + x^2}}\end{aligned}$$

$$\begin{aligned}
\Psi_4 = & -\frac{16\Lambda^2 l^{12} y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{48i\Lambda^2 l^{11} xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{80\Lambda^2 l^{10} x^2 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{112i\Lambda^2 l^9 x^3 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& - \frac{272\Lambda^2 l^8 x^4 y^2}{3(l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8)} + \frac{16i\Lambda^2 l^7 x^5 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{16\Lambda^2 l^6 x^6 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{16i\Lambda^2 l^5 x^7 y^2}{3(l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8)} + \frac{28\Lambda e^2 l^8 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{28\Lambda g^2 l^8 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{12i\Lambda l^9 ny^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& - \frac{72i\Lambda e^2 l^7 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{72i\Lambda g^2 l^7 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{68\Lambda l^8 nxy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{84\Lambda e^2 l^6 x^2 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& - \frac{84\Lambda g^2 l^6 x^2 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{156i\Lambda l^7 nx^2 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{64i\Lambda e^2 l^5 x^3 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{64i\Lambda g^2 l^5 x^3 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{180\Lambda l^6 nx^3 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{20\Lambda e^2 l^4 x^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{20\Lambda g^2 l^4 x^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{100i\Lambda l^5 nx^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{8i\Lambda e^2 l^3 x^5 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{8i\Lambda g^2 l^3 x^5 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{12\Lambda l^4 nx^5 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{4\Lambda e^2 l^2 x^6 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{4\Lambda g^2 l^2 x^6 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{12i\Lambda l^3 nx^6 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{4\Lambda l^2 nx^7 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{12e^4 l^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& - \frac{24e^2 g^2 l^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{12g^4 l^4 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{12i e^2 l^5 ny^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{12i g^2 l^5 ny^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{24i e^4 l^3 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{48i e^2 g^2 l^3 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{24i g^4 l^3 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{60e^2 l^4 nxy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} \\
& + \frac{60g^2 l^4 nxy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} - \frac{24i l^5 n^2 xy^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \frac{12e^4 l^2 x^2 y^2}{l^8 + 4l^6 x^2 + 6l^4 x^4 + 4l^2 x^6 + x^8} + \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"